

Problem Set 6: Bacterial export

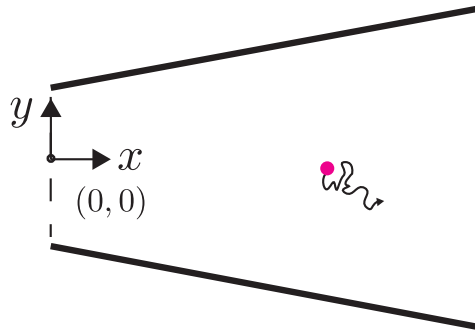
Physics 230A/146A — Biological Physics

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Suppose a molecule is diffusing in a two-dimensional wedge-shaped region with no-flux boundaries

$$y_{\text{top}}(x) = mx + b$$
$$y_{\text{bottom}}(x) = -mx - b$$

for $0 < x < L$, where $m, b, L > 0$ are constants. If the diffusion coefficient is uniform and there are no external forces acting on the molecule, it will tend to move to the right. In this question we ask, can this diffusive motion counteract a left-pulling force? And if so, what is the magnitude of this force?



- The microstates of this system are the (x, y) coordinates of the molecule. Define a macrostate as the set of (x, y) coordinates with a fixed x , i.e., vertical slices. Thus we can label macrostates as x for $0 < x < L$. What is the entropy $S(x)$ of each macrostate x ?
- Assuming no external forces are acting on the molecule — only diffusion — write the free energy $G(x) = -TS(x)$.

- iii. A gradient in an energy functional corresponds to a force. This is true for purely enthalpic energies $E(x)$ and for free energies $G(x)$. What is the force $F(x)$ pulling the molecule? It will depend on x . Which direction does the force pull?

Proteins on the surfaces of cells must be exported from the cell interior. Some unfolded protein in gram-positive bacteria are exported by exploiting entropic forces similar to this problem¹: The polymer is held between a region of high entropy and low entropy, and is effectively sucked into the high-entropy region.

- iv. Assume $b = 2 \text{ nm}$, $L = 100 \text{ nm}$ (the thickness of the gram-positive bacterial cell wall), and $m = 1$. Use the value of $k_B T$ at 30°C (note you need to convert temperature to Kelvins). The force you found in (iii) depends on x . Where is this force largest, and what is its magnitude?

¹Halladin et al., “Entropy-driven translocation of disordered proteins through the Gram-positive bacterial cell wall”, *Nature Microbiology* 2021